

Public or private cloud: Cloud interacts with all the modules, APIs and stores, and it manages as well as processes data. Cloud servers are hosted on the Internet and store data remotely rather than on the local server.

Inventory: This assists in internal provisioning via automation.

API: Systems administrators and developers create APIs for end-to-end interaction of modules.

Modules: Ansible works by connecting to nodes and pushing out small programs, called ‘Ansible modules’ to them. These programs are written as resource models of the desired state of the system. Ansible executes these modules (over SSH by default) and deletes them on finish. Ansible consists of 450 modules for everyday tasks.

Plugins: Plugins are pieces of code that increase the code functionality of Ansible. Ansible is pre-loaded with a number of plugins and also helps the users to write their own. Examples include Action, Cache and Callback.

Networking: Ansible automates diverse networks to make use of all agentless frames and generates useful configurations.

Hosts: These are client machines running Linux or UNIX that come under automation control using Ansible.

CMDB (Configuration Management Database): CMDB is like a repository comprising complete computer networks of IT infrastructure that are fully operational and active for automation of tasks.

Advantages of Ansible:

- Powerful, efficient and highly versatile
- Easy to understand and deploy
- Supports provisioning, application deployment, and orchestration
- Agentless and fully secured

Official website: www.ansible.com

Latest version: 2.10.5

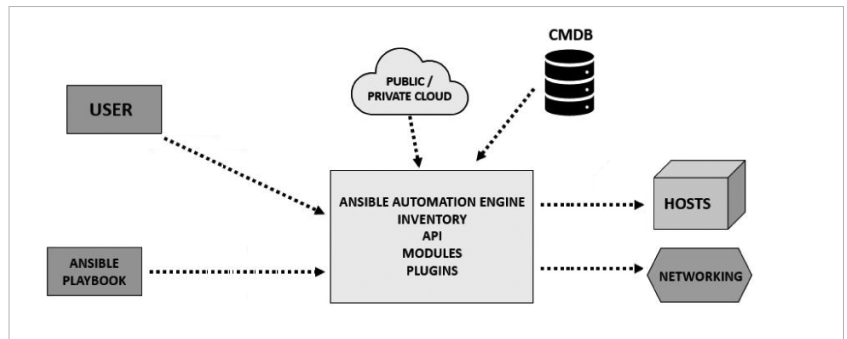


Figure 1: Ansible reference architecture

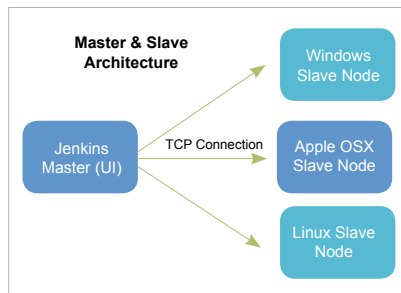


Figure 2: Jenkins reference architecture

Jenkins reference architecture

Jenkins is an open source automation tool used by DevOps teams for bringing continuous integration into projects. It is written in Java and is used throughout the software development process. It not only supports developers in designing, coding and testing but also in deployment by integrating changes to the project in an easy manner. Jenkins is a continuous integration tool, having hundreds of plugins that provide support for automating tasks like building, testing, delivering and deploying the users’ own projects; running tests to detect bugs and other issues, and doing static code analysis, so that users can spend time doing things that machines cannot.

The tool can be accessed and configured using a Web interface, and can be extended with more functionalities through plugins. Jenkins also supports work distribution across several devices, and across diverse platforms.

As a continuous integration tool, Jenkins allows seamless development, testing, and deployment of newly

created code. Continuous integration is a process wherein developers commit changes to source code from a shared repository, and all these changes are built continuously. This can even occur multiple times daily. With every commit continuously monitored by the CI server, code efficiency and verification improve. Jenkins architecture is based on distributed computing. It comprises two components:

1. Jenkins server
2. Jenkins node/slave/build server

Jenkins server: When installed, the Jenkins server can be accessed using a Web interface and run on port 8080. All the jobs and projects are configured using a Web portal dashboard, and build takes place in nodes/slaves. Only one node is configured by default (running on Jenkins’ server), and more nodes can be added using an IP address, user name and password through SSH, jnlp or webstart methods.

The tasks performed by Jenkins server are:

- Scheduling jobs
- Connectivity between nodes and server for execution of tasks
- Monitoring of nodes
- Monitoring of results, and design of overall tasks

Jenkins slave: Slaves work the way they are configured in the Jenkins server, which involves executing build jobs dispatched by the master. Systems administrators can configure the project to always run on particular slave machines or any specific machine, or let Jenkins pick the next available slave/node.